

大物

振动:  $\frac{d^2x}{dt^2} + \omega^2 x = 0 \Rightarrow x = A \cos(\omega t + \varphi)$

$$E_p = \frac{1}{2} k x^2 = \frac{1}{2} k A^2 \cos^2(\omega t + \varphi) \quad \omega = \sqrt{\frac{k}{m}}$$

$$E_k = \frac{1}{2} m v^2 = \frac{1}{2} m \cdot A^2 \omega^2 \sin^2(\omega t + \varphi) = \frac{1}{2} k A^2 \sin^2(\omega t + \varphi)$$

$$E_p + E_k = \frac{1}{2} k A^2$$

合成: ①同方向同频率

$$A = \sqrt{A_1^2 + A_2^2 + 2A_1 A_2 \cos(\varphi_2 - \varphi_1)}$$

$$\tan \varphi = \frac{A_1 \sin \varphi_1 + A_2 \sin \varphi_2}{A_1 \cos \varphi_1 + A_2 \cos \varphi_2}$$

②同方向不同频率

$$\text{"拍"的周期 } T = \frac{2\pi}{\omega_2 - \omega_1}$$

$$\text{拍频 } \nu = \frac{1}{T} = \frac{\omega_2 - \omega_1}{2\pi} = \nu_2 - \nu_1$$

$$\Sigma x = 2A \cdot \cos\left(\frac{\omega_2 - \omega_1}{2} t\right) \cdot \cos\left(\frac{\omega_2 + \omega_1}{2} t + \varphi\right)$$

③垂直, 同频率:

直线/椭圆

④垂直, 不同频率: 李萨如图

$$\text{单摆: } \omega = \sqrt{\frac{g}{l}}, \quad \text{复摆: } \omega = \sqrt{\frac{mgl}{J}}$$

波动:  $dE_p \propto \left(\frac{\partial y}{\partial x}\right)^2$

$$dE_p = dE_k = \frac{1}{2} \rho A^2 \omega^2 \sin^2 \omega \left(t - \frac{x}{u}\right) dV$$

$$\text{能量密度: } w = \frac{dE_p + dE_k}{dV} = \rho A^2 \omega^2 \sin^2 \omega \left(t - \frac{x}{u}\right)$$

$$\bar{w} = \frac{1}{2} \rho A^2 \omega^2$$



平均能流:  $\bar{P} = \bar{w} \cdot u \cdot S = \frac{1}{2} \rho A^2 \omega^2 \cdot u \cdot S$  单位: 瓦

能流密度  $\Rightarrow I = \frac{\bar{P}}{S} = \frac{1}{2} \rho A^2 \omega^2 \cdot u$   
| 波的强度

半波损失: 固定端      自由端无半波损失

干涉:  $\varphi_1 - \varphi_2 = 0$ ,  $\delta = r_2 - r_1 = \begin{cases} \pm k\lambda, \text{加强} \\ (2k+1) \cdot \frac{\lambda}{2}, \text{减弱} \end{cases}$

$\varphi_1 \neq \varphi_2$ , 考虑初相位差.

驻波: 特殊的干涉.      波腹: 干涉加强点  
波节: 干涉减弱点

特例:  $\begin{cases} y_1 = A \cos(\omega t - \frac{2\pi x}{\lambda}) \\ y_2 = A \cos(\omega t + \frac{2\pi x}{\lambda}) \end{cases}$        $y = 2A \cos \frac{2\pi x}{\lambda} \cdot \cos \omega t$

相邻波节之间, 质点相位相同.

※ 一波节两侧, 质点相位相反.

单个质点能量不守恒, 能量在波腹和波节之间转换

波节只有势能, 波腹只有动能

多普勒效应:  $\nu' = \frac{u \pm v_o}{u \mp v_s} \nu$       观察者在分子.  
波源在分母.



## 几何光学

折射:  $n_1 \sin i = n_2 \sin r$

球面镜反射:  $\frac{1}{p} + \frac{1}{p'} = \frac{2}{R}$  (凹正凸负).  $f = \frac{R}{2}$

$$m = \frac{y'}{y} = -\frac{p'}{p}$$

球面折射:  $\frac{n_1}{p} + \frac{n_2}{p'} = \frac{n_2 - n_1}{R}$  (凹负, 凸正).

$$m = \frac{y'}{y} = -\frac{n_1 p'}{n_2 p}$$

薄透镜成像:  $\frac{1}{p} + \frac{1}{p'} = \frac{1}{f}$ , (凹负, 凸正).

$$f = (n-1) \cdot \left( \frac{1}{R_1} - \frac{1}{R_2} \right)$$

$n$  为透镜材质相对环境的折射率

$\Phi = \frac{n_1}{f}$ , ( $n_1$  为透镜所处环境折射率).

## 波动光学

① 双缝干涉:  $\Delta x = \frac{D}{d} \cdot \lambda$

$$x = \pm k \cdot \frac{D\lambda}{d}, (k=0, 1, 2, \dots) \text{明}$$

$$x = \pm (2k-1) \frac{D\lambda}{2d}, (k=1, 2, \dots) \text{暗}$$

$$\delta = n_2 r_2 - n_1 r_1$$

② 等厚干涉:

$$\delta = 2nd + \frac{\lambda}{2}$$

(递增或递减时, 无半波损失).

$$\delta = 2k \cdot \frac{\lambda}{2} \Rightarrow \text{加强}$$

$$\delta = (2k+1) \cdot \frac{\lambda}{2} \Rightarrow \text{减弱}$$

$$\Delta d = \frac{\lambda}{2n}$$

$$l = \frac{\Delta d}{\sin \theta} \approx \frac{\lambda}{2n\theta}$$

③ 牛顿环:

$$\delta = 2nd + \frac{\lambda}{2}$$

$$\delta = 2k \cdot \frac{\lambda}{2} \Rightarrow \text{加强}$$

$$\delta = (2k+1) \cdot \frac{\lambda}{2} \Rightarrow \text{减弱}$$

$$r^2 = R^2 - (R-d)^2 \approx 2Rd$$

$$d = \frac{r^2}{2R}$$



